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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-089

1. Topic (12 words max)

AI-Driven Oral Condition Analysis and Supportive Care Framework

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Oral diseases such as oral cancer, potentially malignant disorders, dental caries, and periodontal disease represent a significant global public health burden, particularly in low- and middle-income countries including Sri Lanka, where vulnerable groups carry a disproportionately high disease load [1]. Oral conditions are among the most common noncommunicable diseases worldwide, affecting an estimated 3.5 billion people, with prevalence rising due to inadequate care access and persistent inequalities in health services [1]. Oral and oropharyngeal cancers are a major component of this burden, with high incidence and mortality globally, especially in populations with prevalent tobacco and betel nut use, necessitating urgent public health intervention [2]. High-risk individuals such as tobacco and betel-quid users, rural residents, elderly persons, and economically disadvantaged communities are most affected, while dental practitioners and primary care providers often face challenges due to subjective assessment methods and limited specialist resources, increasing the disease burden in public health facilities. Socially, untreated oral conditions impair speech, nutrition, social interaction, and psychological well-being; economically, delayed diagnosis leads to high treatment costs and productivity losses; ethically, unequal access exacerbates health disparities; legally, subjective diagnostics increase the risk of misdiagnosis; and environmentally, repeated hospital visits impose additional resource use. Artificial intelligence and deep learning approaches have demonstrated promise in automating early lesion analysis and improving diagnostic support, addressing these limitations and enhancing screening capacity [3].

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The proposed research aligns with Artificial Intelligence in Healthcare, Medical Image Analysis, and Digital Health Systems in the medical technology vertical, and contributes to Sustainable Development Goals by advancing SDG 3 (Good Health and Well-Being), SDG 10 (Reduced Inequalities), and SDG 9 (Industry, Innovation and Infrastructure) through AI-driven preventive and diagnostic innovation [4].

References

[1] World Health Organization, "Oral health," WHO Fact Sheet, 17 Mar 2025. Available: https://www.who.int/news-room/fact-sheets/detail/oral-health

[2] S. Warnakulasuriya, Global epidemiology of oral and oropharyngeal cancer, *Oral Oncology*, vol. 45, no. 4–5, pp. 309-316, Apr.–May 2009. DOI: https://doi.org/10.1016/j.oraloncology.2008.06.002 — Preview: https://pubmed.ncbi.nlm.nih.gov/18804401/

[3] FDI World Dental Federation, "Oral cancer," policy overview, FDI. Available: https://www.fdiworlddental.org/oral-cancer

[4] M. X. et al., "Early Oral Cancer Detection with AI: Design and Implementation of a Deep Learning Image-Based Chatbot," *Applied Sciences*, 2025. DOI: https://doi.org/10.3390/app151910792 — Abstract: https://www.mdpi.com/2076-3417/15/19/10792

6. Brief description of Existing Research and Systems (300 words max)

Existing research in oral health has focused on improving early detection, diagnosis, and management of oral diseases, including oral cancer, dental caries, and periodontal disorders. Traditional systems rely on clinical examination, visual inspection, and tissue analysis, which are often subjective, time-consuming, and dependent on the expertise of dental professionals. Digital imaging tools, such as intraoral cameras, fluorescence-based imaging, and digital radiography, have been developed to enhance lesion visualization and support more accurate clinical decisions.

In recent years, Artificial Intelligence (AI) and deep learning have emerged as promising solutions for automating oral lesion detection and classification. AI models can analyze photographic images of the oral cavity to identify potentially malignant disorders and early-stage cancers with high accuracy in photographic datasets [1][2]. A systematic review and meta-analysis found that AI models achieved diagnostic accuracy ranging widely but often outperform or assist clinical specialists in lesion detection and classification tasks [3]. Some systems are integrated with mobile or web-based platforms, enabling remote screening and triage, which is particularly valuable in rural or underserved communities.

Despite these advancements, existing systems face several limitations. Many AI models are trained on limited datasets, reducing their effectiveness across diverse populations. Most solutions focus on detecting a single disease rather than providing comprehensive oral health assessment.

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Integration into routine clinical workflows remains limited, and ethical, legal, and privacy concerns related to AI-based diagnostics are yet to be fully addressed[2].

The proposed research aims to bridge these gaps by developing an AI-driven framework capable of multi-condition oral screening. By combining deep learning-based image analysis with user-friendly interfaces for both healthcare providers and patients, the system seeks to improve early diagnosis, optimize healthcare resources, and reduce disparities in oral health, contributing to more accessible and preventive oral healthcare.

References

[1] Habal, M. B. (2024). *Automated detection of oral malignant lesions using deep learning: Scoping review and meta-analysis. Oral Diseases*: https://pubmed.ncbi.nlm.nih.gov/39489724/

[2] Sapkal, R., Supnekar, S., Shete, A., Prakash Channe, P., & Desai, A. (2025). *Deep learning and AI in detection of oral malignancy. Journal of Neonatal Surgery*: https://www.jneonatalsurg.com/index.php/jns/article/view/1616?utm_source

[3] Togoo, R. A. et al. (2024). *Enhancing early detection of oral cancer: A comparative study of AI models and clinical specialists. Journal of Oral Medicine*: https://pubmed.ncbi.nlm.nih.gov/41398657/

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The AI-Driven **Oral Condition Analysis and Supportive Care Framework** presents a comprehensive solution for the identified gaps present within the existing oral condition technologies, expounded on within Section 6. Although the present technology shows promising results within the area of early lesion identification as well as the utilization of AI-driven classification techniques, the technology still tends to be individually disease-oriented with strong reliance on expert user analysis as well as the lack of continuous monitoring of the patients. Additionally, many of the present technologies neglect the inclusion of caregivers within the care process.

The proposed framework overcomes these limitations by integrating **multi-condition oral image analysis, two-level AI-based risk scoring, speech therapy monitoring,** and **psychological well-being support with personalized meditation and music therapy** into a single, unified system. Importantly, the system does **not perform direct medical diagnosis,** ensuring ethical safety and regulatory alignment. Instead, it provides **clinically meaningful insights,** supports timely doctor consultations, and automatically **alerts guardians** during high-risk physical or psychological conditions. This holistic integration enhances early intervention, accessibility, and patient-centered care across diverse populations.

System Perspective: Input → Process → Output

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1. Inputs:

- Oral images captured via smartphone or intraoral cameras
- Patient demographic, lifestyle, and clinical history for risk scoring
- Voice samples for speech therapy assessment
- Behavioral and mood data (app usage, sleep, questionnaires)

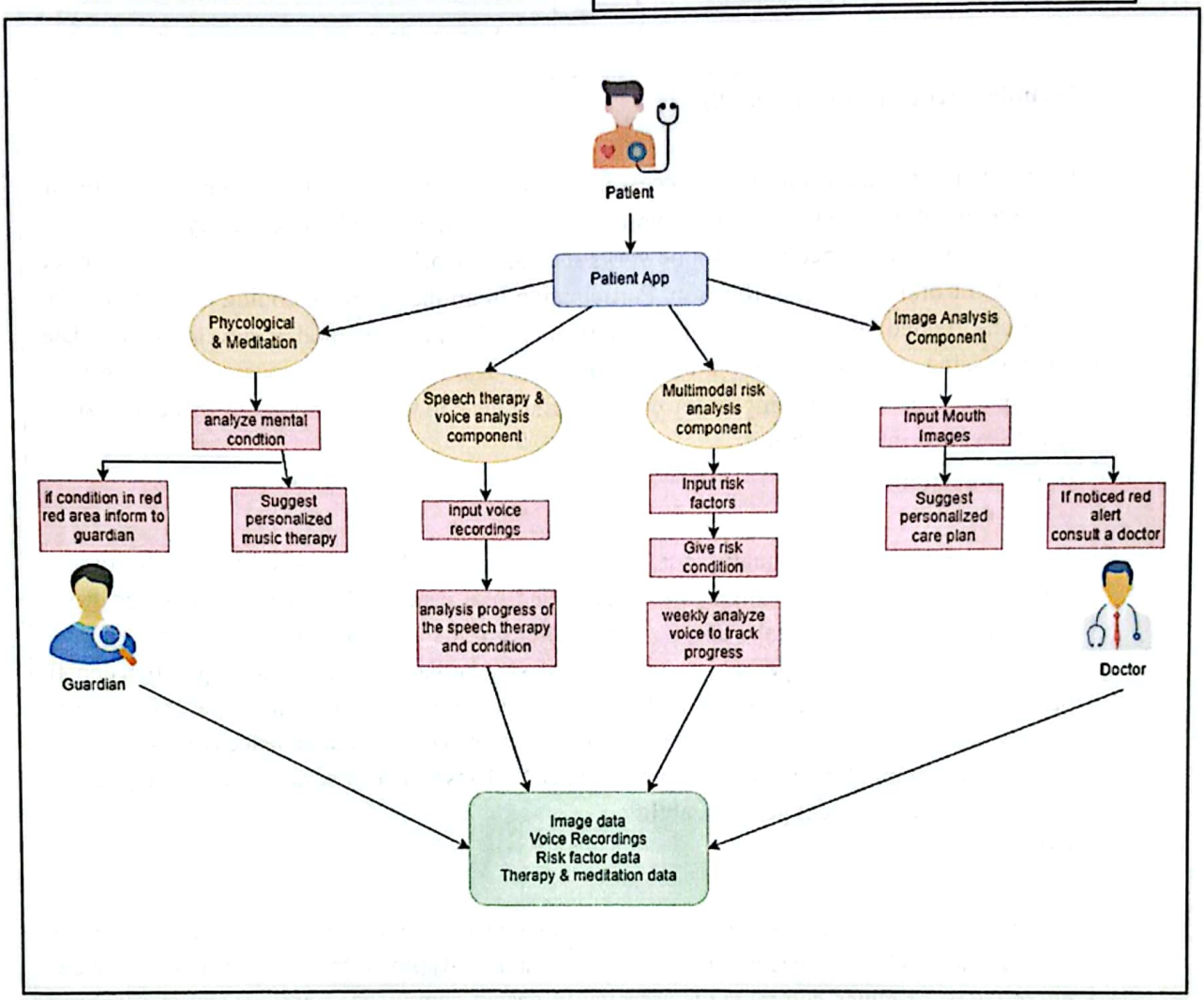
2. Processes:

- **Oral Condition Image Analysis:** A CNN-based AI model analyzes oral images to detect multiple oral conditions and potential abnormalities. The system generates annotated visual reports for clinicians and categorizes severity using color-coded risk levels. When a red-level condition is detected, the system notifies guardians and recommends immediate doctor consultation.
- **Multimodal Risk Scoring Component:** The framework employs a two-tier risk evaluation strategy. Core Risk is computed using clinically validated predictors such as age, smoking, alcohol use, betel chewing, nutrition, and oral hygiene. Extended Risk incorporates personalized modifiers including oral symptoms, lesion indicators, pain, chewing or swallowing difficulty, and voice deterioration trends.
- **Speech Therapy & Voice Analysis:** AI analyzes articulation, clarity, speech consistency, and voice quality from recorded samples, generation of detailed speech therapy progress reports for patients and therapists, supporting rehabilitation planning.
- **Phycological & Meditation Guidance:** The system monitors patient stress and emotional state using behavioral signals and self-reported data. If psychological distress reaches a critical level, guardian alerts are triggered automatically. The AI generates personalized meditation plans and condition-based music therapy, promoting mental relaxation, positivity, and emotional resilience, which play a supportive role in disease recovery.

3. Outputs:

- Clinically annotated oral condition images for healthcare professionals
- Two-level, explainable, and continuously updated oral cancer risk scores
- Personalized patient care plans generated from oral image severity, including recommended actions, oral care guidance, doctor consultation suggestions, and follow-up monitoring
- Speech therapy progress and articulation clarity reports
- Personalized mental health feedback, meditation plans, and music therapy recommendations
- Timely guardian alerts and doctor consultation reminders

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8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful development of the proposed AI-driven oral condition analysis system requires a combination of **specialized dental expertise, technical knowledge in AI and deep learning, and access to high-quality oral health data.**

Specialized Domain Expertise:

Project team members can gain essential expertise in oral health through collaboration with dental professionals, oral pathologists, and clinical researchers. Workshops, online courses, and targeted literature reviews in oral diseases, lesion classification, and preventive dentistry will further enhance understanding of disease pathology and diagnostic protocols. Active engagement with dental clinics and hospitals can provide hands-on exposure to clinical workflows, enabling the team to better align the AI system with real-world needs.

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Technical Knowledge and Skills:

Team members can acquire the necessary technical skills through structured training in machine learning, deep learning, computer vision, and medical image analysis. Practical experience will be gained using frameworks such as TensorFlow or PyTorch, and open-source datasets for oral image classification. Participation in online tutorials, coding bootcamps, and research-oriented workshops will strengthen proficiency in model development, data preprocessing, feature extraction, and performance evaluation. Continuous consultation with AI experts and software engineers ensures adherence to best practices in system design, reliability, and interpretability.

Data Collection:

The system requires high-quality annotated images of oral tissues, including healthy tissue, dental caries, potentially malignant disorders, and oral cancers. Data can be collected from collaborating dental hospitals, clinics, and public health screenings, following standard data acquisition protocols. Images will be anonymized, standardized, and labeled by qualified dental professionals to ensure accuracy. Supplementary patient information, such as age, sex, medical history, and habits like tobacco or betel-quid use, will be collected to enhance risk assessment and model training. Existing open-source oral health datasets may also supplement the primary dataset to improve model generalizability.

Ethical Clearance:

Yes, this project requires ethical clearance, as it involves the collection and use of patient health data, including clinical images and personal information. Approval from an institutional review board (IRB) or ethics committee is essential to ensure compliance with privacy regulations, informed consent procedures, and data protection standards. Adherence to ethical guidelines will guarantee that patient rights are protected while enabling responsible and reproducible research.

By combining **clinical expertise, technical skill development, and rigorous data collection,** the project team is well-positioned to develop a reliable, accurate, and ethically compliant AI system for early detection and risk assessment of oral diseases. Continuous learning, collaboration with domain experts, and adherence to ethical standards will ensure the system meets both clinical and technical objectives.

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9. Objectives and Novelty

Main Objective To develop a holistic oral health mobile system that integrates AI-based oral condition analysis, dual-stage risk assessment, psychological support through guided meditation, and speech therapy monitoring, while delivering patient- centered guidance and guardian-assisted alert mechanisms to ensure timely and appropriate clinical intervention.			
Member Name with Registration No	**Sub Objective**	**Tasks**	**Novelty**
IT22171924 Jeewanthi K H	To analyze mouth images to identify oral condition severity, recommend doctor consultation for high-risk cases, and provide condition-based personalized guidance.	Patients upload images of their mouth through the system. The images are preprocessed and analyzed using image processing and AI techniques to detect visual abnormalities such as color changes, patches, ulcers, and irregular tissue patterns. Based on the analysis, the system classifies the oral condition using color-based severity levels. When a red-level (high-risk) condition is detected, the system alerts the patient, recommends medical consultation, and displays available doctors for further evaluation. Additionally, the system	This component uniquely combines image-based oral condition assessment, color-coded severity indication, automatic doctor recommendation, and condition-specific personalized care planning within a single workflow, while avoiding direct oral cancer diagnosis, which is not commonly addressed together in existing oral cancer screening research.

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		generates a personalized care plan aligned with the detected condition to guide the patient until professional medical care is received.	
Randeniya T D P IT22276278	To develop an AI-based multimodal oral cancer risk analysis and weekly progression monitoring system using lifestyle data, oral health indicators, and voice analysis.	The system first collects the user's demographic and lifestyle information, including age, smoking, betel chewing, alcohol use, nutrition, and oral hygiene, to calculate a baseline oral cancer risk using machine-learning techniques. It then gathers oral health indicators such as ulcers, pain, gum disease, trauma, and swallowing difficulty to refine this into a personalized extended risk score. During initial use, the user records a baseline voice sample representing normal speech. The same sentence is recorded weekly thereafter, allowing the system to extract acoustic features such as pitch, jitter, shimmer, MFCC, and hoarseness related to speech muscle and nerve function. These features are compared with baseline and previous recordings to detect voice changes. A voice-based progression score is generated to indicate improvement, stability, or deterioration.	The key novelty of this system is the use of weekly voice recordings as a digital biomarker for oral cancer progression. Oral and throat tumors often affect nerves and muscles responsible for speech, causing changes such as hoarseness, instability, and slurred voice. By analyzing these changes over time using AI, the system can detect early signs of nerve and tissue involvement that may not be visible in images or symptoms alone. Unlike existing oral cancer apps that only perform one-time image-based detection or questionnaire-based screening, this approach provides continuous, non-invasive, and personalized monitoring. Integrating voice-based progression analysis with lifestyle and oral health risk factors creates a multimodal, time-aware AI framework that improves early

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		Finally, this score is combined with the core and extended risk values to produce a continuously updated oral cancer risk and trend, with alerts generated when high-risk changes are detected.	detection and supports timely medical intervention.
IT22543882 Bakmeewewa H.P.U	To design an AI- based mental health support system that monitors patient stress and depression levels, provides personalized meditation and sound-based stress relief, and alerts a guardian during critical psychological conditions.	The system collects basic psychological and behavioral data such as mood responses, sleep duration, and app usage to calculate stress levels using machine learning. Based on this stress level, users are classified into normal, mild, high, or critical mental risk categories. For normal and moderate stress levels, the system provides personalized meditation, breathing, and relaxation guidance. When high or critical stress is detected, a "Guardian Angel" mechanism is activated, which alerts a registered guardian and automatically starts calming sound or guided relaxation therapy. The system continuously monitors mental well- being over time and adapts support while maintaining ethical use without providing clinical diagnoses.	The novelty of this component is the integration of AI-based psychological risk monitoring with an automatic guardian alert system. Unlike existing systems that focus mainly on physical health, this approach treats mental well- being as a safety-critical factor by combining stress analysis, personalized meditation, and real-time guardian involvement. The "Guardian Angel" mechanism ensures immediate human support during critical depressive states, making mental health care an active and protective feature.

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IT22285706 Arachchige S.S. J	To analyze patient speech, voice, and oral data to evaluate oral cancer progression, classify condition severity, recommend clinician intervention for high- risk cases, and provide personalized care guidance.	Patients upload their speech and voice recordings through the system. The recordings are preprocessed and analyzed using signal processing and AI techniques to detect abnormalities related to oral cancer, such as changes in voice quality, articulation issues, and speech irregularities. Based on the analysis, the system classifies the oral cancer risk using severity levels. When a high-risk (red level) condition is detected, the system alerts the patient, recommends immediate medical consultation, and displays available specialists for further evaluation. Additionally, the system generates a personalized care and monitoring plan aligned with the detected condition to guide the patient until professional medical care is received.	This component uniquely integrates voice and speech signal analysis to monitor therapy progress, severity-level classification using measurable speech features, automated clinician or speech- language therapist recommendations, and condition-specific personalized therapy care planning within a single end-to-end workflow. The system is designed to support therapy monitoring and guidance while intentionally avoiding direct medical diagnosis, an integrated approach that is not commonly addressed together in existing voice and speech therapy research and digital health systems.

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1. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Accepted with minor changes

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Jenny	Krishara	2026/01/12
Co-Supervisor	Dr.	Dinuka	Wijendra	2026/01/12
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor	Dr	Prasanna	Jayasekara	

Summary of external supervisor's (if any) experience and expertise

Dr. Prasanna Jayasekara
BDS (SL), MSc (Com. Dent.), MD (Com. Dent.)
Consultant in Community Dentistry
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Ministry of Health, Sri Lanka.

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